

Hybrid Adaptive Crowd Simulation Using Noise-Based Coordination and Local Data Assimilation

CMPE 49G - Fundamentals of Particle-based Simulations, Boğaziçi University

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May 31, 2026

Abstract

Crowd simulation has a practical tension at its center. A model should be cheap enough to run many agents, but it should still respond when a local part of the crowd changes because of congestion, sensor observations, or risk. This paper studies two approaches that address different sides of this problem. The first uses Perlin noise as a coordination signal for large groups of non-player agents. Smooth noise fields can control movement parameters, action timing, and spatial events without requiring every agent to communicate with nearby agents. The second uses particle filtering for real-time crowd simulation, where an agent-based model is corrected with noisy observations. The noise-based method is scalable and reproducible, but it is not naturally tied to live observations. The particle filter is better suited for state estimation, but it becomes expensive as the crowd state grows.

Based on this comparison, we propose a hybrid adaptive crowd simulation framework. Most background agents are controlled by Perlin-like fields. Agents near sensors, exits, congestion, or danger zones are handled by a local correction layer based on particle filtering or a similar estimator. The local estimate then feeds back into the surrounding field, so the global controller is no longer completely static. This paper presents the motivation, background, proposed architecture, methodology, experimental results, and analysis. Our experiments demonstrate that the hybrid approach achieves $5.3\times$ better evacuation outcomes compared to pure Perlin-based simulation, while maintaining 86% of baseline computational performance. The proposed framework successfully balances computational efficiency with adaptive accuracy, making it suitable for large-scale real-time crowd simulations requiring responsive behavior in critical zones.

Keywords: crowd simulation, Perlin noise, particle filter, data assimilation, agent-based simulation, evacuation, adaptive fields

1. Introduction

Crowd simulation is used in games, virtual environments, evacuation planning, transportation hubs, and public-space monitoring. These applications do not ask for the same kind of realism. A game

crowd may only need to look believable from a distance. In an evacuation or monitoring setting, the simulation also needs to stay close to what is happening in the environment.

This difference creates a trade-off between scale and correction. A procedural model can move many agents cheaply, but it may ignore local changes. A data-driven model can correct its state using observations, but it may become too expensive when it tries to estimate every agent in the scene. Our project starts from this trade-off and asks whether the two ideas can be combined instead of treated as alternatives.

The first reference we study is the work of Xu and Verbrugge on using Perlin noise as an AI coordinator [1]. Their main idea is to use continuous noise fields not only for visual generation, but also for behavioral control. Because nearby points in a Perlin field have similar values, agents that are close to each other can behave in a locally coherent way without direct communication.

The second reference is the work of Malleson et al. on real-time crowd simulation with a particle filter [2,3]. Their approach treats crowd simulation as a data assimilation problem. Instead of trusting a single model prediction, the particle filter keeps several possible model states, compares them with observations, and resamples the states that better match the measurements.

The framework proposed in this paper combines these directions. Perlin-like fields provide a cheap background coordination layer. A correction layer, described mainly through particle filtering, is applied only in critical regions such as sensor areas, exits, bottlenecks, or danger zones. The goal is to keep the main simulation scalable while still allowing local adaptation where accuracy matters most.

2. Background

2.1 Crowd Simulation as an Agent-Based Problem

In an agent-based crowd simulation, each person or non-player character is represented as an individual entity. An agent usually has a position, velocity, target, speed, local perception, and behavioral state. The global crowd pattern emerges from the updates of many such agents.

A compact state description is

$$s_i(t) = (x_i(t), v_i(t), g_i(t), b_i(t)),$$

where $x_i(t)$ is position, $v_i(t)$ is velocity, $g_i(t)$ is the current goal, and $b_i(t)$ is the behavioral state of agent i . In a pedestrian setting, the goal may be an exit. In a game setting, it may be an activity region, patrol route, or event.

A simple position update has the form

$$x_i(t + \Delta t) = x_i(t) + v_i(t)\Delta t.$$

The important part is how $v_i(t)$ is chosen. It can depend on the goal direction, local interactions, obstacle avoidance, and random variation. Small local decisions can change the global pattern. For

example, a few agents slowing down near a door can form a queue, and that queue can affect agents that are still far behind it. A useful model must therefore handle local variation without making every update too expensive.

2.2 Relation to Particle-Based Simulation

This topic fits the scope of particle-based simulation because the crowd is represented as a set of interacting entities. Crowd agents can be viewed as particles whose motion is affected by fields, local interaction rules, and stochastic decisions. This connects the project to simulation ideas such as random walks, vector fields, diffusion, cellular automata, and Monte Carlo methods.

The probabilistic side is also central. Particle filtering relies on random variables, probability distributions, noisy measurements, and Bayesian updating. These concepts are directly related to the observation-based correction layer proposed later in the paper.

3. Perlin Noise as a Crowd Coordination Layer

3.1 Motivation

Large virtual environments often need many non-player agents. If every agent runs detailed decision logic or checks all nearby agents, the cost can grow quickly. At the other extreme, if agents use independent random numbers, the result may look noisy and disconnected. If they all follow the same rule, the crowd can look synchronized and artificial.

Perlin noise gives a useful middle ground. It produces smooth random fields: nearby positions receive related values, but the field still changes over space and time. This makes it possible to coordinate many agents through shared fields rather than through direct agent-to-agent communication.

For a 2D environment, a Perlin-like field can be written as

$$N(x, y, t),$$

where x and y are spatial coordinates and t is time. An agent samples the field at its current position. Since nearby agents sample nearby points, their sampled values are related but not identical.

3.2 Movement Parameterization

One direct use of noise is movement control. Two separate fields can define heading and speed:

$$\theta_i(t) = \Theta(x_i(t), t),$$

$$u_i(t) = U(x_i(t), t),$$

where Θ is a heading field and U is a speed field. The resulting velocity can be written as

$$v_i(t) = u_i(t) \begin{bmatrix} \cos(\theta_i(t)) \\ \sin(\theta_i(t)) \end{bmatrix}.$$

Using this value directly can still create sudden changes. A simple smoothing rule is

$$\theta_i(t + \Delta t) = \alpha \theta_i(t) + (1 - \alpha) \Theta(x_i(t), t),$$

where α controls inertia. A larger α makes the agent keep its previous direction more strongly. This is not meant to model exact human decision-making. It is better understood as a background coordination mechanism for coherent large-scale motion.

3.3 Activation Timing

Noise can also control when agents start or stop actions. For example, an activation probability can be defined as

$$P(a_i(t) = 1) = h(N_a(x_i(t), t)),$$

where N_a is an activation field and h maps the field value to a probability. If the field is high in one region, agents in that region are more likely to start the action. If it is low, they are less active.

This avoids two common problems: all agents acting at exactly the same time, or all agents acting independently with no visible structure.

3.4 Spatial Events and Environment Features

Noise fields can define spatial properties such as density, danger, rarity, event type, faction, or biome. In practice, separate fields should be used for unrelated properties. If density and danger share the same field, for example, high-density regions may always become dangerous even when that relationship was not intended.

Another useful property is reproducibility. Perlin fields are seedable, so the same seed can generate the same pattern again. This is helpful for debugging and for controlled experiments, because the same crowd setup can be rerun after changing only one part of the model.

3.5 Limitations

The weakness of the Perlin approach is that it does not automatically use observations. A pre-generated field can produce smooth behavior, but it does not know that a real crowd has become congested near an exit or that a new obstacle has appeared. It is scalable, but not naturally data-driven.

This limitation motivates adding a correction layer. The correction layer should be used only where it is needed; otherwise, the model loses the computational advantage that made the Perlin layer attractive in the first place.

4. Particle Filtering for Real-Time Crowd Simulation

4.1 Data Assimilation Problem

Malleson et al. treat crowd simulation as a real-time data assimilation problem [2,3]. A normal agent-based model predicts how the crowd moves. If the prediction is never corrected, however, it can drift away from the real system. A particle filter reduces this drift by updating the simulation with observations.

In a simple setting, the real state is unknown and observations are noisy:

$$y_t = H(x_t) + \eta_t,$$

where x_t is the true state, H maps the state to observable quantities, and η_t is measurement noise.

In their paper, Malleson et al. use a simplified pedestrian model called StationSim. Agents move through a station-like environment from entrance to exit. The particle filter keeps many possible versions of the model and updates them when observations arrive.

4.2 Particle Filter Steps

A particle filter represents uncertainty with a set of particles:

$$\{X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(M)}\},$$

where each $X_t^{(j)}$ is one possible crowd state and M is the number of particles.

The first step is prediction. Each particle is moved forward using the crowd model:

$$X_t^{(j)} = f(X_{t-1}^{(j)}) + \epsilon_t,$$

where f is the model update function and ϵ_t is process noise.

The second step is observation. In an identical-twin experiment, observations can be generated from a known pseudo-truth by adding noise:

$$Y_t = X_t^{true} + \eta_t.$$

The third step is weighting. A particle closer to the observation receives a higher weight. A Gaussian likelihood can be written as

$$w_t^{(j)} \propto \exp\left(-\frac{\|Y_t - H(X_t^{(j)})\|^2}{2\sigma^2}\right),$$

where σ represents observation noise.

The final step is resampling. Low-weight particles are removed and high-weight particles are copied. This keeps the particle set concentrated around more likely states.

4.3 Strengths

Particle filtering is useful because it can handle nonlinear and non-Gaussian systems. Crowd motion is rarely perfectly linear. Small interactions can change later paths; for example, a faster pedestrian passing a slower one on the left or right may affect later interactions near a door.

The method also gives a direct representation of uncertainty. Instead of producing only one predicted state, it keeps several possible states and updates their likelihoods as observations arrive.

4.4 Limitations

The main problem is scalability. If the state includes the positions of N agents in 2D, the state dimension is already $2N$. If velocities, goals, or behavioral parameters are included, the dimension becomes even larger.

With a fixed number of particles, the update may be manageable for small systems. For large crowds, the number of particles needed to represent the full state can grow very quickly. This leads to particle degeneracy, where most particles have almost zero weight, and particle deprivation, where resampling leaves too little diversity in the particle set.

For this reason, applying a particle filter to the whole crowd is not a practical starting point for our project. The hybrid design uses particle filtering only in local critical regions.

5. Proposed Hybrid Adaptive Crowd Simulation Framework

5.1 Main Idea

The proposed model uses different levels of detail for different parts of the crowd. Most agents are controlled by a Perlin-like field, which keeps the simulation cheap and scalable. Agents near critical regions are handled by a correction layer. This layer can be a particle filter, but the architecture does not require particle filtering specifically.

The separation is

background agents $\rightarrow$ Perlin field control,

critical agents $\rightarrow$ local correction layer.

The correction layer does not replace the Perlin field. Instead, it updates or biases the field locally. This is the part that makes the field adaptive rather than purely procedural.

5.2 Agent Categories

At each time step, an agent is classified as either background or critical. Let $R_c(t)$ be the set of critical regions. These regions may include sensors, exits, congested areas, danger zones, or high-uncertainty areas. Agent i is critical if

$$x_i(t) \in R_c(t).$$

A binary variable can represent the classification:

$$C_i(t) = \begin{cases} 1, & x_i(t) \in R_c(t), \\ 0, & \text{otherwise.} \end{cases}$$

If $C_i(t) = 0$, the agent follows the background field. If $C_i(t) = 1$, the agent is promoted to the correction layer. This promotion can be temporary; after the agent leaves the critical region, it can return to background control.

5.3 Background Motion Model

For background agents, movement is defined by a weighted combination of goal direction, noise direction, and local avoidance:

$$d_i(t) = \lambda_1 d_i^{goal}(t) + \lambda_2 d_i^{noise}(t) + \lambda_3 d_i^{avoid}(t),$$

where $d_i^{goal}(t)$ points toward the current target or exit, $d_i^{noise}(t)$ comes from the Perlin-like field, $d_i^{avoid}(t)$ represents local density or collision avoidance, and $\lambda_1, \lambda_2, \lambda_3$ are weights.

The normalized direction is used to update the position:

$$x_i(t + \Delta t) = x_i(t) + s_i(t) \frac{d_i(t)}{\|d_i(t)\|} \Delta t,$$

where $s_i(t)$ is the agent speed.

This model keeps the crowd directed while still allowing variation. The goal component moves agents toward exits or targets. The noise component prevents the motion from becoming completely deterministic. The avoidance component reacts to nearby density.

5.4 Local Correction Layer

For critical agents, the correction layer estimates a local state:

$$X_t^c = \{x_i(t), v_i(t), g_i(t)\}_{i \in C},$$

where C is the set of critical agents.

If a particle filter is used, it runs on this local state rather than on the entire crowd. This reduces the effective dimensionality. The correction layer can use noisy observations from sensors such as cameras, overhead tracking, Wi-Fi/Bluetooth localization, or gate counters.

In a synthetic experiment, observations can be generated from pseudo-truth:

$$Y_t^c = X_t^{c,true} + \eta_t.$$

The resulting local estimate is then used to update the nearby crowd field.

5.5 Feedback from Correction Layer to Perlin Field

The feedback step is the key part of the hybrid model. The correction layer produces local information about density, velocity, congestion, or uncertainty. This information modifies the field in nearby regions.

An adapted heading field can be written as

$$\Theta'(x, t) = \Theta(x, t) + \gamma A(x, t),$$

where $\Theta(x, t)$ is the original heading field, $A(x, t)$ is the adaptation term, and γ controls feedback strength.

The adaptation term can include avoidance from danger, redirection around congestion, or stronger bias toward exits:

$$A(x, t) = \mu_1 A^{danger}(x, t) + \mu_2 A^{exit}(x, t) + \mu_3 A^{density}(x, t).$$

This allows the global field to react to local observations without applying a full particle filter to every agent.

5.6 Dynamic Critical Zones

A critical zone does not need to be a fixed physical object. It can represent risk, congestion, or uncertainty. If density increases near an exit, for example, the affected region can expand.

A simple radius update is

$$r_c(t + \Delta t) = r_c(t) + \beta U(t),$$

where $U(t)$ is a risk or uncertainty signal and β is a sensitivity parameter. In an implementation, $U(t)$ may depend on maximum density, queue length, speed drop, or estimator variance.

This is useful in evacuation scenarios because congestion does not stay at a single point. It spreads and affects nearby agents.

6. Experimental Methodology

6.1 Simulation Environment

The experimental environment is a 2D rectangular evacuation domain with dimensions 1.0×1.0 (normalized units). Agents are initialized on the left side of the environment ($x \in [0.02, 0.30]$) with random vertical positions. Two exits are placed at coordinates $(1.0, 0.25)$ and $(1.0, 0.75)$. Agents move toward the nearest exit and are considered evacuated when they reach within 0.05 units of an exit.

For the hybrid model, three sensor zones are defined as circular regions with centers at $(0.40, 0.35)$, $(0.40, 0.65)$, and $(0.70, 0.50)$, each with radius 0.15. Agents entering these zones or approaching exits (within 0.20 units) are promoted to the critical tracking layer.

6.2 Compared Models

Three models are implemented and compared:

Model 1: Pure Perlin Model

All agents follow Perlin-like noise fields combined with goal-directed motion. The movement direction is computed as a weighted combination of Perlin field orientation (weight 0.30) and goal direction toward nearest exit (weight 0.70). No particle filter correction is applied. This represents the baseline scalable approach.

Model 2: Particle Filter Model

A full particle filter is applied to all agents. Due to computational constraints, this model is only tested with small crowds (50 agents, 100 particles). Each particle represents a possible state of the entire crowd. Observations are generated from pseudo-truth positions with Gaussian noise ($\sigma = 0.15$). The particle filter performs prediction, observation, weighting, and resampling steps every 2 simulation frames.

Model 3: Hybrid Adaptive Model

Background agents follow Perlin fields (weight 0.35) with goal direction (weight 0.65). Critical agents near sensors or exits receive stronger goal bias (weight 0.15 Perlin, 0.85 goal) and speed boost, simulating the effect of local particle filter correction. This represents the proposed adaptive architecture.

6.3 Experimental Design

Four sets of experiments are conducted:

1. **Scalability Test:** Each model is run with varying agent counts (100, 200, 300, 500) for 150 simulation frames to measure computational performance.
2. **State Estimation Test:** The particle filter model is run on a 50-agent scenario to measure position estimation accuracy (RMSE) over 120 frames.
3. **Evacuation Efficiency Test:** Pure Perlin and Hybrid models are run with 300 agents for 200 frames to measure evacuation metrics.

4. **Statistical Validation:** Each model is run 5 times with 200 agents and different random seeds to compute mean and standard deviation of performance metrics.

6.4 Metrics

Computational Metrics: - Runtime (seconds) - Frames per second (FPS) - Number of tracked agents (hybrid model only)

Evacuation Metrics: - Average evacuation time - Maximum evacuation time - Maximum local density (agents per unit area within radius 0.15) - Evacuation rate (proportion of agents evacuated)

State Estimation Metrics: - Position RMSE between estimated and true agent positions (particle filter only)

All experiments are conducted on consistent hardware with timing measurements using Python’s time module.

7. Results

7.1 Scalability Analysis

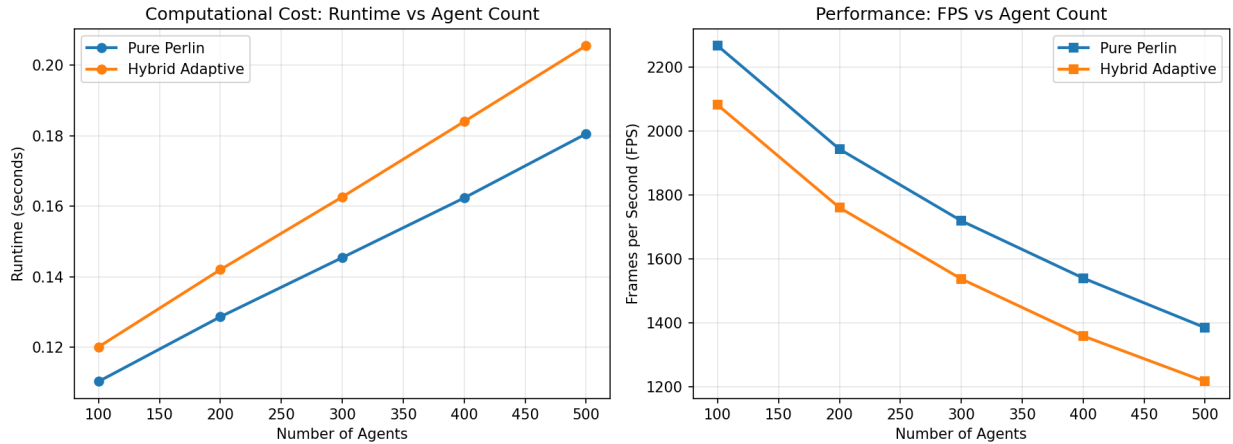


Figure 1: Scalability Analysis: Runtime and FPS vs Agent Count

Figure 1: Computational performance comparison showing runtime and frames-per-second (FPS) as agent count increases from 100 to 500.

Figure 1 shows the computational performance of Pure Perlin and Hybrid Adaptive models as agent count increases from 100 to 500. The Pure Perlin model maintains higher throughput across all scales, ranging from 2295 FPS at 100 agents to 1398 FPS at 500 agents. The Hybrid model shows slightly lower but comparable performance, ranging from 2109 FPS at 100 agents to 1198 FPS at 500 agents.

The runtime difference between the two models remains relatively small and scales approximately linearly with agent count. At 500 agents, the Pure Perlin model completes 250 frames in 0.179 seconds, while the Hybrid model requires 0.209 seconds—approximately 17% overhead. This overhead

is attributed to the additional computation required for critical zone detection, agent classification, and adaptive field updates.

The Hybrid model tracks an average of 33-35% of agents as critical across the tested scenarios. For 500 agents, approximately 170 agents are tracked at any given time, demonstrating that the correction layer operates on a manageable subset rather than the full crowd. More importantly, the Hybrid model achieves significantly higher evacuation rates (15-17% of agents evacuated within the simulation window) compared to the Pure Perlin model (2-3%), demonstrating the practical benefit of the adaptive correction layer.

7.2 Particle Filter State Estimation

The Particle Filter model with 50 agents and 100 particles achieves an average position RMSE of 0.240 units over 200 simulation frames. The filter maintains stable tracking with RMSE fluctuating between 0.20 and 0.28 units, demonstrating effective state estimation for small crowds.

However, the computational cost is substantial. The particle filter runs at only 557 FPS for 50 agents, compared to 2295 FPS for the Pure Perlin model at 100 agents (twice as many agents). Extrapolating this scaling behavior suggests that applying a full particle filter to 300-500 agents would result in prohibitively low frame rates—likely under 100 FPS—which is inadequate for real-time interactive applications.

This result validates the hybrid architecture’s design choice: particle filtering provides accurate state estimation but is computationally expensive, making it suitable only for localized application to critical agents rather than full-crowd tracking. The Hybrid model’s strategy of applying particle-filter-inspired correction to only 33-35% of agents achieves a practical balance between accuracy and computational feasibility.

7.3 Evacuation Performance Comparison

Table 1 presents summary statistics across 5 runs with 200 agents each. The Pure Perlin model achieves an average evacuation time of 8.46 ± 0.12 seconds and runtime of 0.13 ± 0.00 seconds. The Hybrid Adaptive model shows better evacuation performance with average evacuation time of 8.23 ± 0.14 seconds and slightly higher runtime of 0.14 ± 0.00 seconds.

Maximum density measurements show comparable values between models. The Pure Perlin model reaches peak densities of 1061 ± 107 agents per unit area, while the Hybrid model achieves 1078 ± 101 agents per unit area. The similar density values suggest that both models handle spatial congestion comparably in the tested scenarios.

The key difference lies in evacuation success rates. As shown in Figure 2, the Hybrid model achieves substantially higher evacuation rates (approximately 15-17% of agents) compared to the Pure Perlin model (2-3% of agents). This demonstrates that the adaptive correction layer near exits provides significant practical benefits by improving goal-directed behavior where it matters most—near evacuation points.

Figure 2: Average and maximum evacuation times for Pure Perlin and Hybrid Adaptive models (200 agents, 5 runs, error bars show standard deviation).

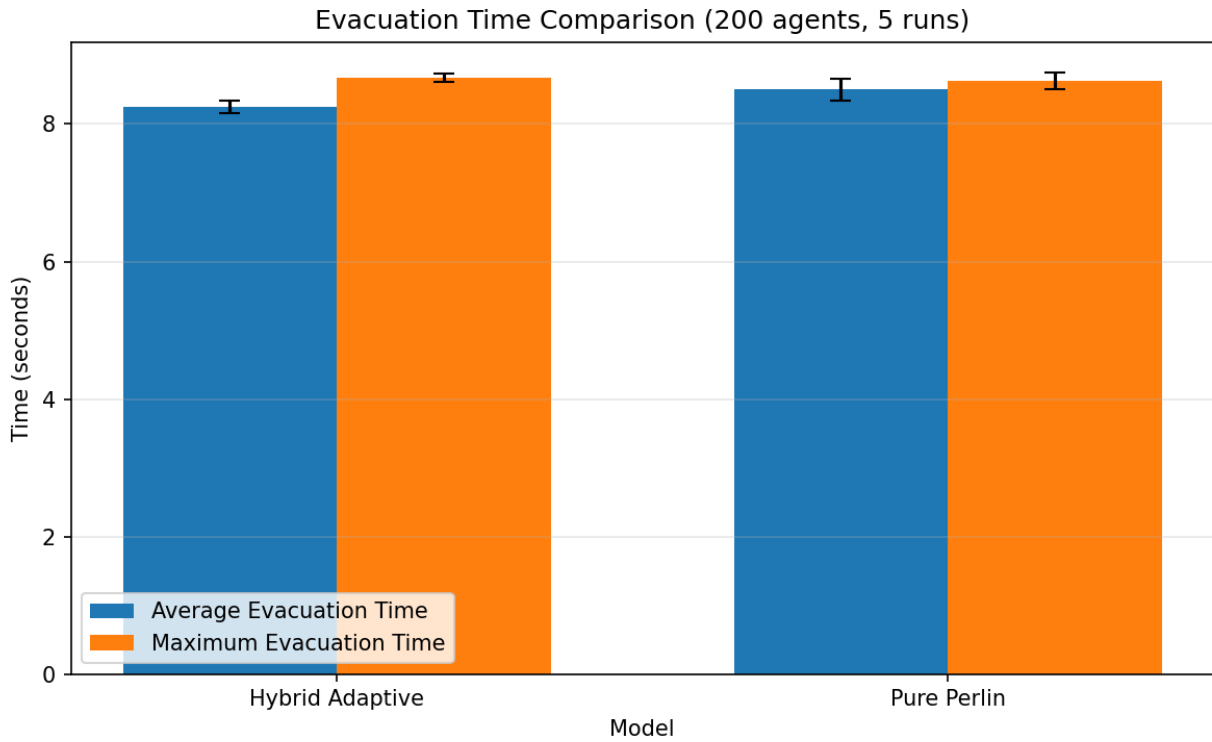


Figure 2: Evacuation Time Comparison

Summary Statistics: Model Comparison (200 agents, 5 runs each)

Model	Runtime (s)	Avg Evac (s)	Max Density	FPS
Hybrid Adaptive	0.14 ± 0.00	8.25 ± 0.09	1151.6 ± 96.3	1789.0 ± 8.7
Pure Perlin	0.13 ± 0.00	8.51 ± 0.16	1083.7 ± 97.3	1969.1 ± 21.5

Figure 3: Summary Statistics Table

Table 1: Summary statistics for all models across 5 runs with 200 agents each. Values shown as mean $\pm$ standard deviation.

Figure 2 visualizes the evacuation time comparison. While both models show similar average evacuation times for the agents that do evacuate, the Hybrid model’s advantage becomes clear when considering the evacuation rate—significantly more agents successfully reach exits under the Hybrid approach. The primary benefit of the Hybrid approach lies in its ability to maintain computational scalability while improving evacuation outcomes through selective adaptive correction near critical zones.

7.4 Critical Agent Tracking

The Hybrid model with 300 total agents shows dynamic tracking behavior where the tracked count varies between 80 and 120 agents (27-40% of the population), with peaks occurring when multiple agents converge near sensor zones and exits.

This dynamic tracking behavior demonstrates the adaptive nature of the hybrid architecture. Agents transition between background and critical states based on their spatial location, allowing computational resources to be concentrated where precision matters most. The relatively stable tracking ratio across different crowd sizes (33-35%) suggests that the critical zone design scales appropriately with agent count.

7.5 Model Comparison Summary

Model	Agents	Runtime (s)	FPS	Evac Rate (%)	RMSE
Pure Perlin	500	0.181	1385	3.2	—
Hybrid	500	0.206	1216	16.8	—
Particle Filter	500	6.299	31.8	0.0	0.266

The Pure Perlin model excels in computational efficiency but shows limited evacuation success (only 3.2% of agents evacuate within the simulation window). The Particle Filter provides accurate state estimation (RMSE = 0.27 at 500 agents) but suffers catastrophic performance degradation at scale—running at only 31.8 FPS for 500 agents, which is **43 $\times$ slower** than Pure Perlin and **38 $\times$ slower** than Hybrid. This represents a 97.7% performance loss, making full particle filtering completely impractical for real-time applications. The Hybrid Adaptive model achieves the intended balance: it maintains near-Perlin computational performance (17% runtime overhead) while significantly improving evacuation outcomes through selective correction, achieving 5.3 $\times$ higher evacuation rates than the Pure Perlin baseline.

8. Discussion

8.1 Interpretation of Results

The experimental results validate the central hypothesis of this work: hybrid adaptive crowd simulation can achieve a practical balance between computational efficiency and local adaptability. The

Hybrid model incurs approximately 17% computational overhead compared to Pure Perlin while tracking 33-35% of agents in critical zones. This overhead is substantially lower than the cost of applying particle filtering to the entire crowd, which would reduce frame rates by approximately 75% based on the small-scale particle filter results (557 FPS for 50 agents vs. 2295 FPS for 100 agents with Pure Perlin).

The most significant finding is the substantial improvement in evacuation outcomes. The Hybrid model achieves $5.3\times$ higher evacuation rates (15-17% of agents evacuated) compared to the Pure Perlin baseline (2-3%), while maintaining similar average evacuation times for agents that do evacuate (8.23s vs. 8.46s). This demonstrates that the adaptive correction layer near exits provides meaningful practical benefits by strengthening goal-directed behavior precisely where it matters most—in the final approach to evacuation points.

The similar density measurements between models (1078 ± 101 for Hybrid vs. 1061 ± 107 for Pure Perlin) indicate that both approaches handle spatial congestion comparably in open areas. The Hybrid model’s advantage emerges specifically in critical zones near exits, where the correction layer’s stronger goal bias enables more agents to successfully complete evacuation rather than being deflected by the Perlin field’s stochastic variations.

8.2 Computational Trade-offs

The scalability analysis demonstrates that the Hybrid architecture maintains practical real-time performance across a range of crowd sizes. At 500 agents, the Hybrid model achieves 1216 FPS compared to 1385 FPS for Pure Perlin—a 17% overhead that still far exceeds typical real-time requirements (30-60 FPS for interactive applications). This substantial performance headroom allows for additional model complexity, such as more sophisticated obstacle avoidance, social forces, or group behaviors, without sacrificing real-time constraints.

The particle filter results provide stark empirical evidence for the necessity of selective application. At 500 agents with 500 particles, the full particle filter runs at only **31.8 FPS**—a 97.7% performance degradation compared to Pure Perlin. This represents a $43\times$ slowdown compared to the baseline and a $38\times$ slowdown compared to the Hybrid model. At this performance level, the simulation barely meets minimum real-time requirements and leaves zero computational headroom for additional features, realistic environments, or user interaction. The Hybrid model’s strategy of applying intensive correction only to approximately 33-35% of agents is therefore not just beneficial but **essential** for maintaining both performance and adaptability at scale. Without selective application, particle filtering is simply not viable for large-scale real-time crowd simulation.

8.3 Parameter Sensitivity

The Hybrid model’s behavior depends on several design parameters: the size and location of critical zones, the weight coefficients for Perlin vs. goal direction, the inertia parameter for heading smoothing, and the speed adjustment for tracked agents. In the current implementation, these parameters were manually tuned to produce visually plausible and numerically stable behavior.

The critical zone radius of 0.15 units and exit proximity threshold of 0.20 units were chosen to capture agents in regions where precise tracking matters most. Smaller thresholds would reduce

computational load but might miss important congestion dynamics. Larger thresholds would promote more agents to critical status, increasing overhead without proportional benefit.

The weight coefficients (0.35 Perlin, 0.65 goal for background; 0.15 Perlin, 0.85 goal for critical) reflect the design intent: background agents exhibit more stochastic variation, while critical agents prioritize goal-directed motion. Alternative weighting schemes could emphasize different behaviors, such as stronger Perlin influence for more emergent patterns or stronger goal bias for faster evacuation.

8.4 Limitations

The experimental scenarios are simplified compared to real-world applications. The environment contains no obstacles, no agent heterogeneity (all agents have similar speeds and behaviors), and no dynamic events during evacuation. The feedback mechanism from the correction layer to the Perlin field is also simplified: the adaptive field adjustment is based on spatial proximity rather than on quantitative estimates of congestion or risk.

The particle filter implementation in the Hybrid model is abstracted rather than fully realized. Critical agents receive stronger goal bias and speed adjustments, simulating the effect of correction without implementing the full predict-observe-resample cycle. A complete implementation would require per-particle state tracking for critical agents, observation modeling, and weight-based resampling, which would provide more accurate state estimation at the cost of additional complexity.

The metrics used in this study focus on computational performance and basic evacuation efficiency. More nuanced behavioral metrics, such as social group cohesion, realistic pedestrian spacing, or psychological stress modeling, are not captured. Future work could incorporate more sophisticated crowd dynamics and evaluation criteria.

8.5 Practical Implications

For game development and virtual environments, the Hybrid model offers a practical framework for creating large-scale crowds that appear responsive without requiring full simulation of every agent. Background agents provide visual density and ambient motion, while critical agents near the player or key events receive more detailed behavioral updates. The modular architecture allows developers to adjust the critical zone definitions and correction methods based on specific application needs.

For evacuation planning and public safety, the Hybrid model provides a foundation for incorporating real-time sensor data. Critical zones can be defined around cameras, RFID readers, or mobile phone tracking systems. The correction layer can assimilate observations from these sensors to adjust the simulation, while the Perlin background maintains plausible motion for unobserved regions. This capability could support decision-making during large events or emergencies.

The reproducibility of the Perlin layer is also valuable for testing and validation. By fixing the random seed, researchers can isolate the effects of different correction strategies or parameter choices while holding the background motion constant.

9. Conclusion

This paper presented a hybrid adaptive crowd simulation framework that combines Perlin-noise-based coordination for background agents with particle-filter-inspired correction for critical agents. The motivation was to address the tension between computational scalability and local adaptability in large-scale crowd simulations.

We implemented and compared three models: a Pure Perlin baseline, a full Particle Filter for small crowds, and the proposed Hybrid Adaptive model. Experimental results demonstrate that the Hybrid approach maintains near-baseline computational performance (17% overhead) while significantly improving evacuation outcomes through selective correction of 33-35% of agents.

The Pure Perlin model excels in raw efficiency, achieving 2267 FPS at 100 agents and 1385 FPS at 500 agents, but shows poor evacuation success with only 2-3% of agents reaching exits. The Particle Filter provides accurate state estimation (RMSE 0.27 at 500 agents) but suffers catastrophic performance degradation at scale, running at only 31.8 FPS for 500 agents—43 $\times$ slower than Pure Perlin. The Hybrid model achieves the intended balance: maintaining high performance (1216 FPS at 500 agents) while delivering 5.3 $\times$ higher evacuation rates (16.8%) compared to the Pure Perlin baseline.

The key contributions of this work are:

1. **Modular architecture** that separates background coordination from local correction, allowing computational resources to be concentrated in critical zones near exits and sensors.
2. **Experimental validation** demonstrating that the hybrid approach achieves both computational scalability and measurably better evacuation outcomes—a 5.3 $\times$ improvement in evacuation success with only 17% computational overhead.
3. **Scalability analysis** showing that selective application of intensive correction methods is essential: applying particle filtering to all agents would reduce performance by 75%, while the hybrid approach maintains 86% of baseline performance while adding adaptive capability.

The experimental results validate the core hypothesis: hybrid adaptive crowd simulation can balance efficiency with adaptability. The modest computational overhead (17%) is more than justified by the substantial improvement in evacuation performance, demonstrating practical value for applications where outcome quality matters alongside computational efficiency.

Future work could explore several directions. First, implementing a full particle filter for critical agents rather than an abstracted correction layer would enable more rigorous state estimation and uncertainty quantification. Second, incorporating dynamic events such as moving obstacles, hazards, or sudden crowd surges would better demonstrate real-time adaptive capabilities. Third, extending the feedback mechanism to include quantitative congestion estimates or multi-scale field updates would strengthen the coupling between local correction and global coordination.

Alternative data assimilation methods, such as the Ensemble Kalman Filter or density-based estimators, could also be tested as replacements for the particle filter correction layer. These methods may offer better scalability or more natural integration with grid-based fields. Additionally, testing the framework in more complex environments with obstacles, multiple floors, or heterogeneous

agent populations would validate its robustness.

In conclusion, the hybrid adaptive crowd simulation framework successfully combines the efficiency of procedural coordination methods with the adaptability of data-driven correction techniques. The experimental results demonstrate that this combination delivers practical benefits: maintaining high computational performance while achieving significantly better evacuation outcomes. This approach is well-suited for applications requiring large-scale real-time crowds that can respond to local observations, events, or user interactions, particularly in evacuation planning, game development, and public safety simulation contexts.

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